Coffee Shop Network Setup Documentation

Prepared October 28, 2026. Configuration work logged between 11:58 AM and 12:44 PM.

1. Overview

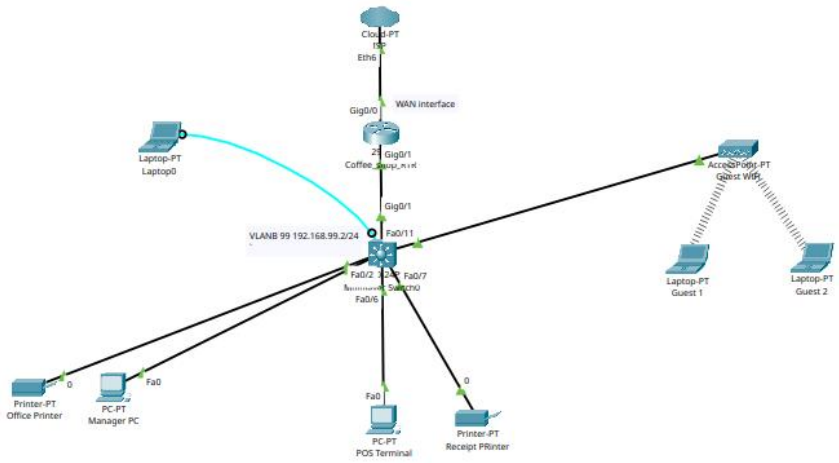
This document describes the small network I built for a coffee shop. The goal was simple: keep the business side of the shop (the manager's computer, the office printer, and the point-of-sale gear) safe and separate, while still giving customers free guest Wi-Fi.

To do that, the network is split into four VLANs. A multilayer switch connects every device in the shop, and a single router sits between the switch and the internet provider. The router handles all traffic moving between VLANs, which is a setup commonly called "router on a stick." Guests can reach the internet, but they are blocked from touching anything in the office, POS, or management networks.

Everything was built and tested in Cisco Packet Tracer.

2. Network Diagram

The picture below shows how everything is physically connected. The ISP cloud connects to the router, the router connects to the switch, and every shop device plugs into the switch. The light blue line from Laptop0 is a console cable that was used to configure the switch.



3. Devices in the Network

Device	Hostname / Label	Role
Cloud-PT	ISP	Stands in for the internet provider
Router	CoffeeShop-RTR	Connects the shop to the ISP and routes between VLANs
Multilayer switch	CoffeeShop-MLS	Core switch that every device connects to

Device	Hostname / Label	Role
Access point	Guest WiFi	Provides wireless access for customers
Laptop-PT	Guest 1	Customer laptop on guest Wi-Fi
Laptop-PT	Guest 2	Customer laptop on guest Wi-Fi
PC-PT	Manager PC	Manager's workstation
Printer-PT	Office Printer	Printer for the back office
PC-PT	POS Terminal	Cash register / point-of-sale computer
Printer-PT	Receipt Printer	Prints customer receipts
Laptop-PT	Laptop0	Admin laptop, connected by console cable for setup

4. Physical Connections and Ports

Here is exactly what plugs into what.

From	Port	To	Port	Notes
ISP Cloud	Eth6	CoffeeShop-RTR	Gig0/0	WAN interface
CoffeeShop-RTR	Gig0/1	CoffeeShop-MLS	Gig0/1	Trunk link carrying all VLANs
CoffeeShop-MLS	Fa0/1 - Fa0/2	Manager PC and Office Printer	Fa0 / 0	Management Office VLAN
CoffeeShop-MLS	Fa0/6	POS Terminal	Fa0	POS VLAN
CoffeeShop-MLS	Fa0/7	Receipt Printer	0	POS VLAN
CoffeeShop-MLS	Fa0/11	Guest WiFi access point	Port 0	Guest VLAN
Laptop0	RS232	CoffeeShop-MLS	Console	Used for configuration only

Beyond the devices that are drawn, I reserved extra ports so the shop can grow later. Ports Fa0/1 to Fa0/5 are set aside for the office, and Fa0/6 to Fa0/10 are set aside for POS equipment.

5. VLANs and IP Addressing

Each VLAN gets its own subnet, and the router holds the .1 address in each one as the gateway.

VLAN ID	Name	Subnet	Gateway (Router)	Who uses it
10	MANAGEMENT_OFFICE	192.168.10.0/24	192.168.10.1	Manager PC, Office Printer
20	POS	192.168.20.0/24	192.168.20.1	POS Terminal, Receipt Printer
30	GUEST_WIFI	192.168.30.0/24	192.168.30.1	Guest access point and customer devices
99	NETWORK_MANAGEMENT	192.168.99.0/24	192.168.99.1	Switch management

The switch itself has a management address on VLAN 99: **192.168.99.2/24**, with its default gateway set to **192.168.99.1** (the router).

6. Switch Configuration (CoffeeShop-MLS)

Basic setup and security

I started by giving the switch a proper name and locking it down:

- Hostname set to CoffeeShop-MLS
- DNS lookup turned off so typos in commands don't cause long delays
- Enable secret password set, and password encryption turned on so passwords don't show in plain text
- Login banner added warning that access is for authorized people only
- Console line protected with a password, with synchronous logging so messages don't interrupt typing

Remote access with SSH

Instead of Telnet, I set up SSH so the switch can be managed securely from the network:

- Domain name set to coffeeshop.local
- Local admin account created with full privileges
- RSA encryption keys generated and SSH version 2 enforced
- All 16 virtual terminal lines (VTY 0 to 15) use local login, accept SSH only, and time out after 10 minutes of inactivity

Access ports

Every port that connects to an end device is set to access mode and uses PortFast, which lets devices come online faster. PortFast is only used on ports that connect to end devices, never to other switches.

Ports	VLAN	Purpose
Fa0/1 - Fa0/5	10 (MANAGEMENT_OFFICE)	Office devices

Ports	VLAN	Purpose
Fa0/6 - Fa0/10	20 (POS)	Point-of-sale devices
Fa0/11	30 (GUEST_WIFI)	Guest Wi-Fi access point

Trunk to the router

Gig0/1 is the link to the router. It runs as an 802.1Q trunk that only allows VLANs 10, 20, 30, and 99. Limiting the allowed VLANs keeps unwanted traffic off the link.

7. Router Configuration (CoffeeShop-RTR)

Basic setup and security

The router got the same treatment as the switch: a hostname, DNS lookup disabled, an enable secret, password encryption, a warning banner, and a protected console line.

Interfaces

Interface	Connects to	Configuration
Gig0/0	ISP (Eth6)	Turned on, described as the ISP connection
Gig0/1	Switch (Gig0/1)	Turned on, split into four sub-interfaces below
Gig0/1.10	VLAN 10	192.168.10.1/24, encapsulation dot1Q 10
Gig0/1.20	VLAN 20	192.168.20.1/24, encapsulation dot1Q 20
Gig0/1.30	VLAN 30	192.168.30.1/24, encapsulation dot1Q 30
Gig0/1.99	VLAN 99	192.168.99.1/24, encapsulation dot1Q 99

The sub-interfaces are what let one physical cable act as the gateway for four different networks.

DHCP

The router hands out IP addresses automatically, so nobody has to set them by hand. The first 20 addresses in each range are excluded and kept free for devices that need a fixed address, such as printers or the access point.

Pool name	Network	Default gateway	Excluded addresses
Management_office_pool	192.168.10.0/24	192.168.10.1	.1 to .20
POS_pool	192.168.20.0/24	192.168.20.1	.1 to .20
Guest_wifi_pool	192.168.30.0/24	192.168.30.1	.1 to .20

8. Guest Network Restrictions (ACL)

This is the most important security piece. Customers should be able to get online, but they should never be able to reach the shop's own computers, the POS system, or the network equipment. I created an extended access list named **GUEST_RESTRICTIONS** and applied it inbound on the guest sub-interface (Gig0/1.30).

The rules, in order:

1. Allow DHCP requests, so guest devices can still get an address
2. Deny guests from reaching the Management Office network (192.168.10.0/24)
3. Deny guests from reaching the POS network (192.168.20.0/24)
4. Deny guests from reaching the Network Management network (192.168.99.0/24)
5. Allow everything else from the guest network, which means internet access

Because the router reads the list from top to bottom, the deny rules catch internal traffic first, and whatever is left over is allowed out to the internet.

9. Verification

After each stage I ran checks to make sure things worked. These are the commands I used and what each one confirmed:

Command	What it checks
show vlan brief	VLANs exist and the right ports are in them
show interface brief	Ports are up and in the expected mode
show ip interface brief	IP addresses are assigned and interfaces are up
show ip dhcp pool	DHCP pools are configured correctly
show access-list	The guest ACL rules are in place and counting matches
show run	The full running configuration, saved with write memory

I saved the configuration after almost every change with the write command so nothing would be lost if a device restarted.

10. Things to Improve Next

The setup works, but looking back over the configuration log, here are a few things worth cleaning up or finishing:

- **DNS servers.** All three DHCP pools use 0.0.0.0 as the DNS server, which is a placeholder. Real DNS addresses (for example, the ISP's or a public DNS service) should be added so devices can resolve website names.
- **Internet routing.** The log doesn't show an IP address on the router's WAN port (Gig0/0), a default route, or NAT. Those will be needed for the shop's devices to truly reach the internet.
- **Shared password.** The same password is used for the console, enable secret, and admin account. Using different, stronger passwords for each would be safer.

- **Spelling clean-up.** A few descriptions and banners contain typos (for example "Unauathorized", "mangement", and "Assingned"). They don't affect how anything works, but they're worth fixing before handing this over.
- **Switch default gateway.** The switch uses a default gateway for its own management traffic, which is correct for a switch acting at Layer 2. If routing is ever enabled on the switch, this will need a second look.
- **Testing.** Ping tests between VLANs and from the guest laptops should be run and recorded to prove the ACL blocks and allows the right traffic.

11. Summary

In short, the coffee shop now has a clean, organized network: four VLANs, one router doing the routing, automatic IP addresses for every device, secure remote management over SSH, and a guest network that is walled off from everything important. With the few improvements above, it will be ready for real-world use.